

WELL COMPLETION REPORT (SYNTHETIC)

OIL-DGB-009 - Digboi-009

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000097
Field / District	Digboi / Tinsukia
Block	PEL-ASSAM-01
Basin	Assam-Arakan
Formation	Girujan (GRJ)
Spud / Completion	2005-03-01 / 2005-04-29
Measured / TVD	2155 m / 2106 m
Trajectory	Vertical; inc 2.5 deg; az 125.8 deg
Formation Interval	1304-1806 m
Lithology	Silty shale with sandstone lenses
Pore Pressure Reference	2274 psi
Fracture Gradient	14.09 ppg
Mud System	Low-Solids WBM
NWIS Well Risk	Low (32.9/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	538	Surface
9 5/8 in casing	1249	Intermediate
7 in liner/casing	1896	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2005-03-16 682 m	Pressure Spike	Critical	Pressure trend exceeded recent moving baseline. Mitigation: Condition hole and confirm pressure normalization. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.
2005-03-15 646 m	Mud Loss	High	Active pit volume declined during stable pump rate. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Track flow balance and pit volume trend before losses become severe.
2005-03-28 1100 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2005-03-29 1117 m	Kick	High	Flow-out exceeded flow-in and gas units increased. Mitigation: Re-establish overbalance and monitor pit/flow balance. Lesson: Maintain mud-weight window using offset pressure evidence.
2005-03-30 1151 m	Torque Spike	High	Sharp torque excursion occurred with RPM instability. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2005-03-31 1185 m	Stuck Pipe	High	Hookload became erratic during pipe movement. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Avoid prolonged static periods in high-drag interval.
2005-04-25 2010 m	NPT	High	Drilling parameters were stable but operations were suspended. Mitigation: Verify equipment health before resuming drilling. Lesson: Separate geological risk from equipment-driven NPT in analytics.

Completion Summary

The synthetic well reached 2155 m MD in the Girujan interval. The NWIS knowledge record links 12 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.